

Codetantra Practical 1

Practice Lab Assignment

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1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$
where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).
Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

Test case 1

5.0

18.0

50.00kgm/s

Test case 2

2.3

4.8

11.04kgm/s

calculate...

```
1 mass=float(input())
2 velo=float(input())
3 mom=mass*velo
4 print(f"{mom:.2f}kgm/s")
5
```

Average time

0.015 s

15.25 ms

Maximum time

0.018 s

18.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

18 ms

Debug

Expected output	Actual output
5.0	5.0
18.0	18.0
50.00kgm/s	50.00kgm/s

Test case 2

15 ms

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

Test case 1

9

81

Test case 2

100

10.00

Test case 3

24

4.90

Test case 4

condition...

```
1 import math
2 n=int(input())
3 if 1<=n<=9 :
4     print(n**2)
5 elif 10<=n<=99:
6     print(f"{math.sqrt(n):.2f}")
7 elif 99<=n<=999:
8     result=n**(1/3)
9     print(f"{result:.2f}")
10 else :
11     print("Invalid")
```

Average time0.011 s11.14 ms

Maximum time0.013 s13.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 143 ms

Expected output9

Actual output81

Test case 242 ms

Expected output100

Actual output10.00

Test case 3403 ms

Expected output24

Actual output4.90

Test case 4

Expected output

Actual output

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

Test case 1

2014-07-02

2014-11-02

123

Test case 2

2014-07-02

2014-07-11

9

noOfDays...

```
1 from datetime import datetime
2 date_str1=input()
3 date_str2=input()
4 date_format= "%Y-%m-%d"
5 d1=datetime.strptime(date_str1,date_format)
6 d2=datetime.strptime(date_str2,date_format)
7 delta=d2-d1
8 print(delta.days)
```

Average time

0.032 s

32.00 ms

Maximum time

0.008 s

88.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2

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1.1.4. Reverse a Number

22/55

reversen...

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Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

Test case 1

5367

7635

Test case 2

8432

231

Explorer

1
2
3
4

n=int(input())
reversed_num=int(str(n)[::-1])
print(reversed_num)

Debugger

Average time
0.012 s
12.40 ms

Maximum time
0.015 s
15.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1

15 ms

Debug

Expected output

5367

7635

Actual output

5367

7635

Test case 2

12 ms

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
`<number> x <multiplier> = <result>`

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

Test case 1

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

8 x 6 = 48

8 x 7 = 56

8 x 8 = 64

Explorer

multiplica...

1 #write your code here...
2 n=int(input())
3 for i in range(1,11):
4 print(f"{n} x {i} = {n*i}")
5
6

Average time
0.008 s
8.25 ms

Maximum time
0.010 s
10.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 10 ms

Debug

Expected output

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

Test case 1
4 55 85 78 89
Aggregate Percentage: 71.75 Grade: First Division

Test case 2
5

passorfa...

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

n=int(input())
marks=list(map(int,input().split()))
passed_all=True
if len(marks) !=n:
 print("Invalid input")
else:
 if any (mark<40 for mark in marks):
 print("Fail")
 else:
 percent=sum(marks)/n
 print(f"Aggregate Percentage: {percent:.2f}")
 if percent>75:
 print("Grade: Distinction")
 elif 60<percent<75:
 print("Grade: First Division")
 elif 50<percent<60:
 print("Grade: Second Division")
 elif 40<percent<50:
 print("Grade: Third Division")

Average time: 0.016 s
Maximum time: 0.025 s
5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1
Expected output: 4
Actual output: 4
55 85 78 89
Aggregate Percentage: 71.75
Grade: First Division

Test case 2
Expected output: 5
Actual output: 5
Aggregate Percentage: 71.75
Grade: First Division

Terminal Test Cases

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12.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

Test case 1

0 1 1 2 3

Test case 2

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377

fibonacci...

```
1 def fibonacci(n):
2     if n==1:
3         return 0
4     elif n==2:
5         return 1
6     else:
7         return fibonacci(n-1)+fibonacci(n-2)
8     # return fibonacci( )
9 n = int(input())
10 for i in range(1, n+1):
11     print(fibonacci(i), end=" ")
12
```

Average time0.043 s43.25 msMaximum time0.129 s129.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output0 1 1 2 3Actual output0 1 1 2 3

Test case 2

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

Test case 1

5

*
* *
* * *
* * * *
* * * * *

Test case 2

4

*
* *
* * *
* * * *

rightangl...

```
1 n=int(input())
2 for i in range(1, n +1):
3     for j in range(i):
4         print("*", end=" ")
5     print()
```

Average time0.009 s9.50 ms

Maximum time0.011 s11.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

Expected output

5

*
* *
* * *
* * * *

Actual output

5

*
* *
* * *
* * * *

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

Test case 1

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Test case 2

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

numberP...

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1):
4         print(j+1,end=" ")
5     print()
6
```

Average time: 0.021 s, 21.00 ms

Maximum time: 0.035 s, 35.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Actual output

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Terminal

Test cases

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